Lecture 11: Infinite Series

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what an infinite series is and how it is formed from a sequence.
- ☐ Determine whether a series converges or diverges using partial sums.
- ☐ Recognize and work with geometric series, including finding their sums.
- ☐ Understand the nth-term test and apply it even if the question does not suggest it.
- ☐ Know how the manipulation of the series affecting its convergence behaviour.

Introduction

Consider the sequence $\{a_1, a_2, a_3, \ldots, a_n, \ldots\}$.

Now, let define another sequence

$$\{s_n\} = \{a_1, a_1 + a_2, a_1 + a_2 + a_3, \ldots, a_1 + a_2 + a_3 + \cdots + a_n, \ldots\}$$

As n grows larger, what is happening to this sequence $\{s_n\}$?

Infinite Series

Given a sequence of numbers $\{a_n\}$, an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

is an infinite series. The number a_n is called the nth term of the series. The sequence $\{s_n\}$ defined by

$$\begin{aligned} s_1 &= a_1 \\ s_2 &= a_1 + a_2 \\ &\vdots \\ s_n &= a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k \\ &\vdots \end{aligned}$$

is the sequence of partial sums of the series, and s_n is the nth partial sum.

Convergence and Divergence

If sequence $\{s_n\}$ converges to a number L, we say that the series converges and that its sum is L. In this case, we also write

$$a_1 + a_2 + \cdots + a_n + \cdots = \sum_{n=1}^{\infty} a_n = L.$$

If $\{s_n\}$ does not converge, we say that the series diverges.

Example

Determine whether $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}}$ converges or diverges, if it converges, evaluate the infinite series.

Geometric Series

Geometric series are series of the form

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$$

where $a, r \in \mathbb{R}$ and $a \neq 0$.

A geometric series diverges if $|r| \geq 1$.

If $|r| < 1$, the geometric series converges and

$$\sum_{n=1}^{\infty} ar^{n-1} = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r} = \frac{a}{1-r}.$$

Example

1. $a = \frac{1}{9}$ and $r = \frac{1}{3}$

2. $\sum_{n=0}^{\infty} \frac{(-1)^n 5}{4^n}$

Example (Telescoping Series)

Determine whether $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges or diverges, if it converges, evaluate the infinite series.

The nth-Term Test for Divergence

Considering the series

$$\sum_{n=1}^{\infty} \frac{n+1}{n}$$

The nth-Term Test for Divergence

Theorem: If $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \to 0$.

The nth-Term Test for Divergence

The nth-Test for Divergence

$\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n\to\infty} a_n$ fails to exist or is different to zero.

The nth-Term Test for Divergence

Examples:

1. $\sum_{n=1}^{\infty} n^2$
2. $\sum_{n=1}^{\infty} \frac{n+1}{n}$
3. $\sum_{n=1}^{\infty} (-1)^{n+1}$
4. $\sum_{n=1}^{\infty} \frac{-n}{2n+5}$

The nth-Term Test for Divergence

Example: Consider the series

$$1+\frac{1}{2}+\frac{1}{2}+\frac{1}{4}+\frac{1}{4}+\frac{1}{4}+\frac{1}{4}+\cdots+\frac{1}{2^n}+\frac{1}{2^n}+\cdots+\frac{1}{2^n}+\cdots$$

Combining Series

Theorem: If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

1. Sum Rule: $\sum(a_n + b_n) = \sum a_n + \sum b_n = A + B$
2. Difference Rule: $\sum(a_n - b_n) = \sum a_n - \sum b_n = A - B$
3. Constant Multiple Rule: $\sum ka_n = k\sum a_n = kA$ for any number k

Combining Series

As corollaries of previous theorem:

1. Every nonzero constant multiple of a divergent series diverges.
2. If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum(a_n + b_n)$ and $\sum(a_n - b_n)$ both diverge.

Combining Series

Examples:

(a) $\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}}$

(b) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{4}{2^n}$

Adding or Deleting Terms

Adding or deleting finite terms to or from a series will not affect the series's convergence or divergence. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$.

Conversely, if $\sum_{n=k}^{\infty} a_n$ converges for any $k > 1$, then $\sum_{n=1}^{\infty} a_n$ converges.

Reindexing

As long as the order of its terms is preserved, reindexing any series will not alter its convergence.